

A Novel Method of Skeletonization of Complex Shapes Based on Bisectors

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The mathematical skeleton of a complex form has been essential for a variety of scientific fields and of great interest to many researchers for decades. It is based on several concepts such as the reconstruction of forms and image processing. This paper aims to develop a novel mathematical algorithm to approximate the skeleton of a non-polygonal shape and to compare it to the most used methods. The mathematical technique of skeletonization is used as a reference to validate and compare the proposed method to the most used ones. The crux of the proposed technique is to Cartesianize the shape (polygonize in 2D), then skeletonize it. Moreover, this novel method is grounded upon the construction of bisectors on the simplex of the corresponding Cartesianized shape. Python is used to implement the algorithm proposed and test it on multiple shapes. The comparison of the results generated by the proposed algorithm and the Python predefined function `skeletonize()` shows that the proposed method is more effective and could be adjusted through the rate of Cartesianization of the target shape. The major contributions of this novel technique include the mitigation of some issues of existing methods, simplification, and optimization of the processing performance mainly in terms of algorithm complexity.

Keywords: Skeleton; exposed points; bisectors; polygonization; Cartesianization.

1. Introduction

Certainly, skeletons are reedy and locally centered structures, which reflect the geometrical, topological, and symmetrical characteristics of shapes efficiently and instinctively such as in Fig. 26. Skeletons have been a powerful tool for shape segmentation,¹ manipulation,^{2,3} and blending.⁴ Multiple methods have been proposed, which can be divided mainly into two classes. The first class includes methods that

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generate surface skeletons,^{5,6} the second class encompasses methods that generate curve skeletons.^{7,8} While the first class captures shape geometry much better, the second class is faster, simpler, and the most adopted in applications.⁹ To differentiate between the quality of skeletons generated by the curve skeletonization technique, multiple criteria are used such as centeredness, reediness, compactness, noise resistance, homotopicity, and computational cost.¹⁰ Multiple challenges have emerged from the existing methods. For instance, Mesh rigging has been a challenge to skeletonization, as generated skeletons show many defects such as shrinkage of end branches and escaping the original input shape boundaries, as shown in Fig. 1. Furthermore, other methods have exhibited issues concerning the quality criteria for curve skeletonization.¹⁰ To overcome these issues, manual intervention might be required, which is error-prone and time demanding. Rigging¹² and shape segmentation¹ for instance require high-quality curve skeletons, which have been provided so far by improving skeletonization methods. Proposed new approaches might present new issues, and might need additional adjustable parameters, or have a more complex implementation.⁹ In addition, they have to be exhaustively tested on wide collections of shapes.^{10,13} Another issue with the majority of skeletonization approaches is that they cannot guarantee that the same level of detail is preserved on identical areas of the input shape. This presents an opportunity for further research and investigations.

The skeleton has many applications, especially in the field of recognition of shapes and imagery such as in Refs. 14 and 15. Moreover, other applications can be derived from these techniques such as finding the shortest path that links all border points, which could be used for emergency response and criminal prosecution within a city or as cover techniques for a cancer region to optimize the time of radiotherapy.

The complex forms are still difficult to control and study, given the complexity of their borders, see Fig. 1. To simplify this complexity, it is necessary to find a method or a few methods to Cartesianize (polygonize) these shapes. For the sake of simplification of geometrical structures, one approach is to discretize the borders to facilitate the approximation of the skeleton of the targeted region. In this study, an

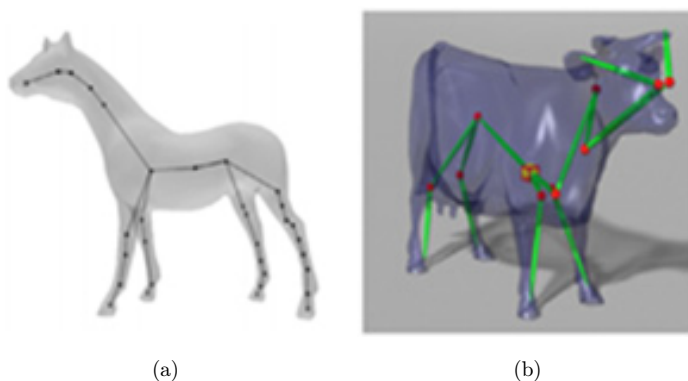


Fig. 1. Skeletons generated in Ref. 11 (a) and Ref. 2 (b) using mesh rigging.

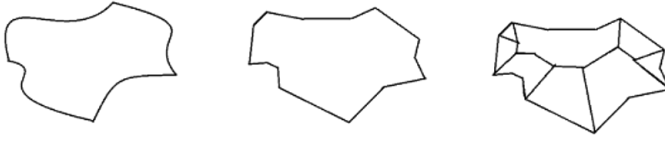


Fig. 2. The complex region on the left and the corresponding polygon in the middle, the skeleton profile on the right.

algorithmic technique is proposed, which is based on bisector construction. The ultimate goal of this paper is to Cartesianize a complex two-dimensional (2D) shape, and then to skeletonize it, see Fig. 2. This method does not require complex mathematical manipulations as it is purely geometric. Furthermore, the algorithm proposed is flexible, and efficient, hence will accelerate the speed of both convergence and consistency. While some skeletonization algorithms process continuous geometric data, others process discrete a only. However, these two representations can be changed using well-known algorithms: voxelization can turn continuous geometric data into a discrete representation,¹⁶ while surface extraction can turn voxelized data into a geometric representation.¹⁷ The quality of the approximation depends on the discretization rate. That is, the more Cartesianized the shape is, the more precise the skeleton is. Consequently, the degree of approximation is called the degree of Cartesianization. Subsequently, this degree induces the degree of skeletonization.

The bisectors technique could be used in 2D and three-dimensional (3D) spaces, it mitigates the issues that might be hindering other methods such as rigging as shown in Fig. 1, where skeletons are not locally centered and, in some instances, leak out of the boundaries of the shape.

All of these methods have to be validated by the original mathematical definition given by maximal balls (3D) or maximal disks (2D), which is developed in the preliminaries and definitions section.

2. Preliminaries and Definitions

This section introduces preliminaries and definitions that are necessary to establish the framework of the study.

Definition 1. Let us define a hypersphere centered on x $B(x, r)$ according to the norms by

$$B(x, r) = \{y \in R^n : \|x - y\| \leq r\}.$$

Therefore, particularly if $n = 3, n = 2$, we have

The ball: $B(x, r) = \{y \in R^3 : \|x - y\| \leq r\}$,

The disk: $D(x, r) = \{y \in R^2 : \|x - y\| \leq r\}$.

Definition 2. Let R be a region included in R^3 . The boundary of R is defined as

$$\partial R = \text{Adh}(R) \setminus \text{int}(R).$$

$\text{Adh}(R)$ is the adherence of R and $\text{int}(R)$ is the interior of R .

Definition 3. A ball $B(x, r)$, respectively, a disk $D(x, r)$ is said to be maximal in R if there are less than three points, respectively, two points of tangency in R^{18} (see Fig. 3).

With the last two definitions above, let us define for each $x \in R$, the minimal distance to the boundary of R as $\delta(x, \partial R) = \min_{y \in \partial R} \|x - y\|$. This definition will be used in the coming sections. This distance is minimal to the boundary as schematized in Fig. 4, which is the distance from x following the normal direction to the boundary. Now, the definition of the skeleton of R represented by $\text{Sk}(R)$ is as follows.

Definition 4. If R is in R^2

$$\text{Sk}(R) = \left\{ x \in R, \exists x_1 \neq x_2 \in \partial R : \|x - x_1\| = \|x - x_2\| = \min_{y \in \partial R} \|x - y\| \right\}$$

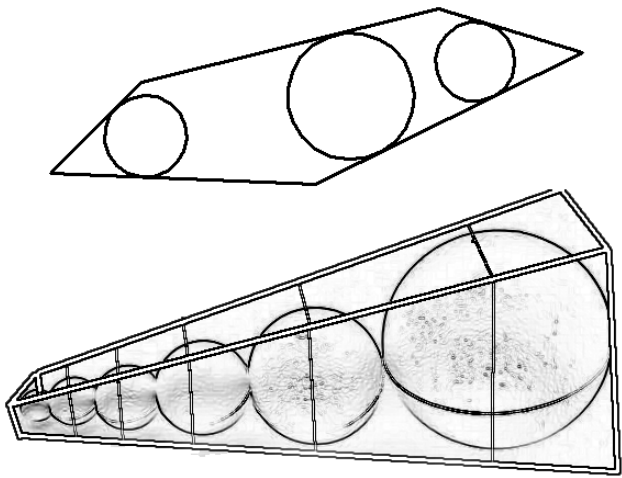


Fig. 3. The disk or balls inside the Cartesianized shape are maximal.

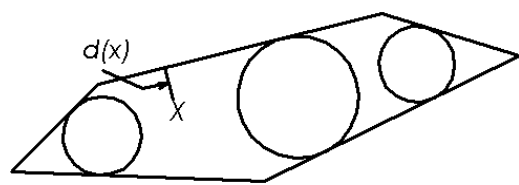


Fig. 4. The minimal distance from x to the boundary.

If R is in R^3

$$\begin{aligned} \text{Sk}(R) &= \left\{ x \in R, \exists x_1 \neq x_2, \neq x_3 \in \partial R : \|x - x_1\| \right. \\ &= \|x - x_2\| = \|x - x_3\| \\ &= \left. \min_{y \in \partial R} \|x - y\| \right\}. \end{aligned}$$

Theorem 5. *The skeleton of a region R included in R^2 , respectively, included in R^3 is the set of all centers of maximal disks, respectively, maximal balls of the region R .*

Proof. The reader is referred to Rousseau.¹⁸ $\square$

According to Definition 4 and Theorem 5, skeleton $\text{Sk}(R)$ is defined as the set of maximal ball centers. For instance, Fig. 5 shows the construction of the skeleton of an elliptic shape, using the same definition.¹⁸

Another experimental method for the calculation of the skeleton is inspired by the following idea: The skeleton is the set of points where the lines of fire cross perpendicularly to the boundary and finish at the fire intersections.¹⁸ However, this method presents three fundamental problems¹⁸:

- The first one assumes that the curve of ∂R is a function in $C^1[R^2]$ or $C^1[R^3]$, which means that the curve is continuously differentiable. That is not available in the case of a polygon with singular points;
- The second problem is that classical algorithms are heavy to program, as they have to compute the gradient at several points of the border;
- The third one is that we cannot easily find the function of the boundary, especially when the borders are crooked. For the details of the algorithm, refer to Ref. 18.

The proposed technique will not compute many points, but only some nodes and branches. Before giving the algorithm, it is necessary to introduce some lemmas and

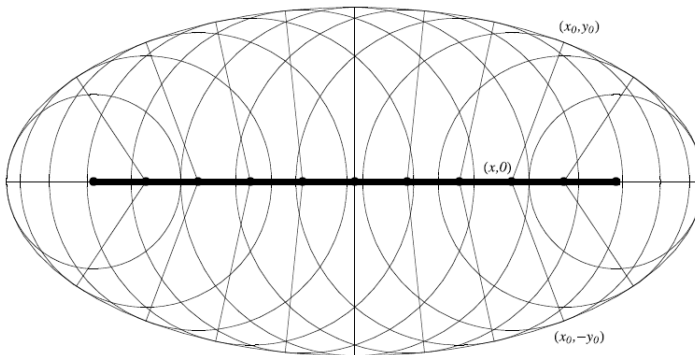


Fig. 5. The skeleton of an ellipse is the line segment in the middle: The set of all maximal balls' centers.¹⁸

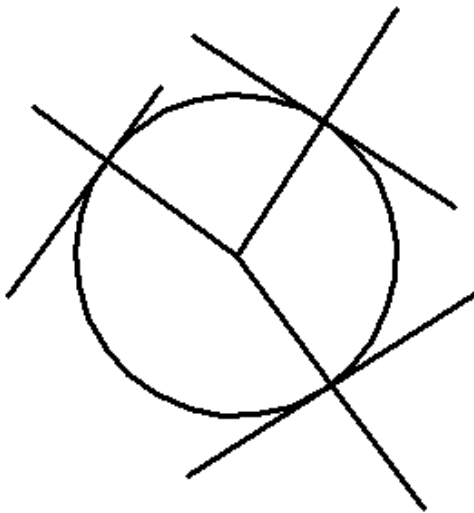


Fig. 6. The normal vectors to the circle contour pass through the center.

propositions to demonstrate that the bisectors method is equivalent to the Skeleton definition.

Lemma 6. *A line directed by the normal vector at any point in the boundary of a hypersphere passes through the center of the hypersphere, as presented in Fig. 6 for a disk.*

Proof. The reader is referred to Euclidean geometry.¹⁹ □

Proposition 7. *Let R be a region, then, for each point X of ∂R , there exists a maximal ball or a disk maximal tangent to ∂R at X , as presented in Fig. 7.*

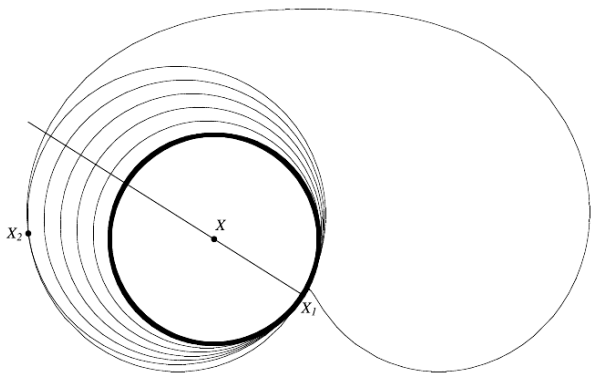


Fig. 7. Step-by-step construction of the maximal disk related to a point X_1 of ∂R .

Proof. Let $X_1 \in \partial R$ and N are the normal line to the ∂R at X_1 . Let be a point $X \in N$ which belongs to the $\text{int}(R)$. Consider the disk $D(X, r_1)$ with $r_1 = d(X, X_1)$. $D(X, r_1)$ is tangent to ∂R at X_1 . Then, exchange X is replaced by another center $X \in N$ following the way to the interior of R . If $D(X, r_1)$ is tangent to the ∂R at another point X_2 , then the process is stopped. If not, the technique continues until finding a center $X \in N$ such that $D(X, r_n)$ with $r_n = d(X, X_2)$ is tangent to ∂R at X_2 . Therefore, $D(X, r_n)$ is maximal in R and tangent to ∂R at X_1 and X_2 . This technique is called the blowing technique. $\square$

Remark 8. It can be confirmed that the normal vectors to the boundary ∂R meet the skeleton $\text{Sk}(R)$ because for each point X of ∂R there exists a maximal disk or a maximal ball which has a tangency at X .

The next sections discuss the geometrical analysis that is necessary to build the skeleton of a polygonal shape in R^2 and R^3 .

3. Skeletonization of Polygons

Lemma 9. *Given two concurrent segments of the boundary of a shape R ; the corresponding skeleton is formed of the bisector of the angle related to the intersection of these segments. See Fig. 8.*

Proof. In this demonstration, the study is restricted to the R^2 case. For the R^3 space case, it would be the same technique, the bisector segments should be replaced by the bisector plans and the intersection points should be replaced by the intersection segments. $\square$

Let R be a region formed of two concurrent segments $[OA]$ and $[OB]$ as shown in Fig. 8. Let's prove that $\text{Sk}(R) = H$, with H is the bisector segment of the angle between the two segments. So, it is to prove that $\text{Sk}(R) \subset H$ and $H \subset \text{Sk}(R)$. Let $X_0 \in \text{Sk}(R)$. According to Theorem 5, there exists a maximal disk centered at X_0 and tangent to ∂R at X_1 and X_2 . In this case, it is required to prove that the angle $\widehat{X_0OA} = \widehat{X_0OB}$. Since $X_1 \in [OA]$ and, $X_2 \in [OB]$, then

$$\widehat{X_0OX_1} = \widehat{X_0OA}$$

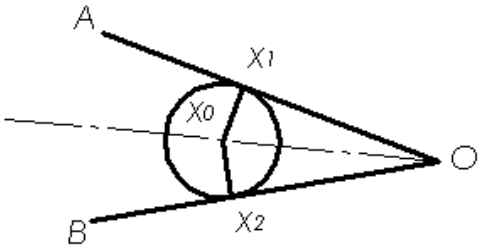


Fig. 8. (OX_0) : segment bisector for angle AOB.

and

$$\widehat{X_0 O X_2} = \widehat{X_0 O B};$$

$d(X_0, X_1) = d(X_0, X_2)$ because these distances correspond to the radius of the maximal disk. Therefore, since X_1 and X_2 belong to the normal line, then

$$\widehat{X_0 X_1 O} = \widehat{X_0 X_2 O} = \frac{\pi}{2}.$$

Thus, it can be concluded that the triangles $\widehat{O X_0 X_1}$ and $\widehat{O X_0 X_2}$ are rectangles and have two equal common segments. So, their hypotenuses are equal.

Hence

$$\widehat{X_0 O X_1} = \widehat{X_0 O X_2} \iff \widehat{X_0 O A} = \widehat{X_0 O B},$$

and, $X_0 \in H$, the bisector of the angle $\hat{O}$.

For the reverse inclusion, let $X_0 \in H$, and $X_1 \in [OA]$ such that the segment $[X_0 X_1]$ is perpendicular to $[OA]$, and let $X_2 \in [OB]$ such as $[X_0 X_2]$ is perpendicular to $[OB]$. The triangles $O X_0 X_1$ and $O X_0 X_2$ are equilateral because

- (1) The angle $\widehat{O X_0 X_1} = \widehat{O X_0 X_2} : X_0 \in H$,
- (2) $\widehat{O X_1 X_0} = \widehat{O X_2 X_0} = \frac{\pi}{2}$.

Then, $d(X_0, X_1) = d(X_0, X_2)$;

Then, $d(X_0, X_1) = \min_{y \in [OA]} \{d(X_0, y)\}$.

Let $C(X_0, d)$ be the circle tangent to $[OB]$ at X_1 . So, $C(X_0, d)$ passes by X_1 , and then passes by X_2 because $d = d(X_0, X_1)$ is the radius of the circle. Therefore, X_0 is the center of the maximal circle (maximal disk) of the region R according to Theorem 5, so that

$$X_0 \in \text{Sk}(R).$$

In conclusion, $\text{Sk}(R) \subset H$ and $H \subset \text{Sk}(R)$, therefore

$$\text{Sk}(R) = H.$$

Similarly, Lemma 10 gives the same result when the shape R is formed by two parallel segments.

Lemma 10. *The skeleton of a shape R , which is formed of two parallel segments $[AA']$ and $[BB']$ are the medium segments (medial axis) $[MM']$ of the two segments as it is presented in Fig. 9.*

Proof. Let $X_0 \in [MM']$, and $X_1 \in [AA']$ such that $[X_0 X_1]$ is perpendicular to $[AA']$. It is clear that $d(X_0, X_1) = \min_{y \in [AA']} d(X_0, y) = d$. Similarly, let X_2 be a point of the segment $[BB']$ such that X_0 is the medium of $[X_1 X_2]$. This is possible because X_0 belongs to the medium line. So, the circle $C(X_0, d)$ tangent to $[AA']$ at X_1 is tangent to $[BB']$ at X_2 . Hence, the disk is maximal at the interior of $[AA']$ and $[BB']$, which means that $[MM'] \subset \text{Sk}(R)$. The reverse inclusion is obvious. $\square$

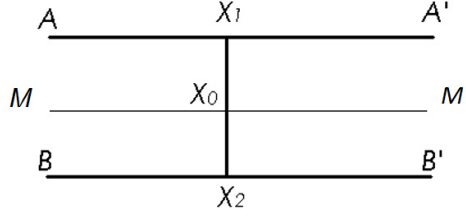


Fig. 9. Segment bisector as a median line of two parallel segments.

Lemma 11. *Let two concurrent straight lines have respective equations $D_1 : a_1x + b_1y + c_1 = 0$ and $D_2 : a_2x + b_2y + c_2 = 0$. Then, the bisector B is described by the following equation²⁰:*

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}. \quad (1)$$

Now, let's present the different results that are used as tools for skeletonizing a given Cartesianized shape. In the next sections, the authors present the two sub-algorithms that correspond to

- The Cartesianization (discretization) of a continuous shape;
- The skeletonization of the discretized shape.

4. 2D Cartesianization and Skeletonization Algorithm

Begin Algorithm (Bisectors)

Step I: Cartesianization

The Cartesianization is a newly generalized concept of polygonization in R^2 and polytopes in R^n , that has been introduced in Ref. 21. The Cartesianization is established in infinite-dimensional spaces such as Banach spaces. The first step is to construct an approximate profile of the region R with the linear borders, as described in Fig. 3. The degree of accuracy of the method depends on the number of discretization points and the curvature radius.

(a) **Initialization:** A given level of curvature $C > 0$ is fixed.

(b) **Considering a set of discretized points:** $D = \{P_i; i = 1, \dots, n\}$ in the boundary of the analyzed region R as described in Fig. 10.

Choice of a set of vertices: (External nodes) of the boundary:

Let E be a set of vertices of a local region R ;

For each triplet (P_{i-1}, P_i, P_{i+1}) , the radius of the curvature C_i is computed by the following equation:

$$r_i = \sqrt{(x_i - x_{c_i})^2 + (y_i - y_{c_i})^2}, \quad (2)$$

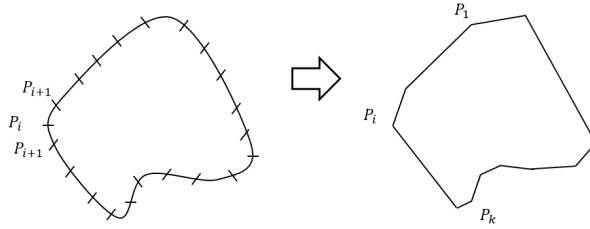


Fig. 10. Discretized region in the left and its Cartesianized profile P following the fixed radius of curvature and boundary orientation.

where (x_i, y_i) are the coordinates of the point P_i and (x_{c_i}, y_{c_i}) are the coordinates of the center O_i of the tangent circle to the curvature C_i . The geometrical parameters of the curvature C_i are detailed in Fig. 10.

So, (x_{c_i}, y_{c_i}) are the coordinates of the center O_i of the circle passing through the points $\{P_{i-1}, P_i, P_{i+1}\}$.

The coordinates (x_{c_i}, y_{c_i}) are computed using the following equations:

$$x_{c_i} = \frac{\frac{x_{i+1}^2 - x_i^2 + y_{i+1}^2 - y_i^2}{2(y_{i+1} - y_i)} - \frac{x_i^2 - x_{i-1}^2 + y_i^2 - y_{i-1}^2}{2(y_i - y_{i-1})}}{\frac{x_i - x_{i-1}}{y_i - y_{i-1}} - \frac{x_{i+1} - x_i}{y_{i+1} - y_i}}, \quad (3)$$

and

$$y_{c_i} = -\frac{x_i - x_{i-1}}{y_i - y_{i-1}} x_{c_i} + \frac{x_i^2 - x_{i-1}^2 + y_i^2 - y_{i-1}^2}{2(y_{i+1} - y_i)}. \quad (4)$$

(d) **Test:** If $r_i \geq C$ then P_i is put in E . The points are called external nodes or vertex M_i . Else, P_i is rejected and E is incremented by 1.

The algorithm continues with the same previous steps by the next triplet (P_i, P_{i+1}, P_{i+2}) .

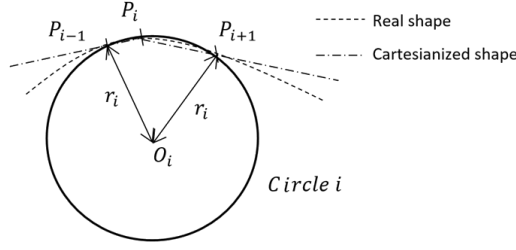
This sub-algorithm is stopped until it reaches a maximal number of point $\epsilon(e)$

The vertices that belong to the set E are connected by lines: The Cartesianized profile is finally obtained.

Step II: Skeletonization

With the results above, let's present the algorithm of skeletonization of the Cartesianized profile P in R^2 , which is illustrated in Fig. 10. The main purpose of this technique is to determine the skeleton nodes, and then connect them. Identification of the nodes is the most delicate phase of this algorithm. Before discussing the algorithm, some notations should be defined as follows:

- (1) Bisectors are supposed to be the beams of fire moving to the interior of the polygon P ;
- (2) M_i : external node (vertex);
- (3) N_i : internal node;


 Fig. 11. Geometric parameters at a curvature C_i .

- (4) Prioritized bisector: Bisectors, that meet each other first, that is, they intersect inside P before any other intersection, as shown in Fig. 12;
- (5) Each angle necessarily corresponds to a node.

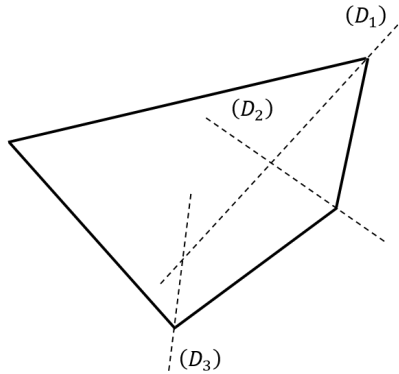
Step I: The first step of the algorithm is to compute the bisector of two neighboring angles with the same segment and define their point of intersection (Fig. 13).

M_{i-1} , M_i , M_{i+1} , M_{i+2} are four neighboring vertices belonging to the polygon P that is defined in a previous step of discretization. Let (x_i, y_i) be the coordinates of M_i . Then, the equations of the lines $(D_{i-1,i}) = \frac{(M_{i-1}M_i)}{\|M_{i-1}M_i\|}$ and $(D_{i,i+1}) = \frac{(M_iM_{i+1})}{\|M_iM_{i+1}\|}$ are defined by the following systems:

$$\begin{cases} (D_{i-1,i}) : a_{i-1,i}x + b_{i-1,i}y + c_{i-1,i} = 0, \\ (D_{i,i+1}) : a_{i,i+1}x + b_{i,i+1}y + c_{i,i+1} = 0, \end{cases} \quad (5)$$

s.t.

$$\begin{cases} a_{i-1,i} = \frac{y_i - y_{i-1}}{x_i - x_{i-1}}, \\ b_{i-1,i} = -1, \\ c_{i-1,i} = y_{i-1} - \frac{y_i - y_{i-1}}{x_i - x_{i-1}} x_{i-1}. \end{cases} \quad (6)$$


 Fig. 12. D_1 and D_2 are prioritized, but not with D_3 because D_1 meets first D_2 .

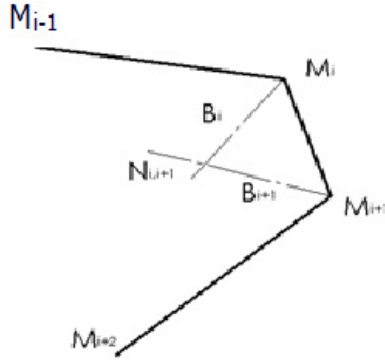


Fig. 13. The intersection of two bisector segments is called an internal node.

Similarly, for the segment $(D_{i,i+1})$, $i - 1$ is replaced by i and i by $i + 1$. The geometric parameters of the line $(D_{i,i+1})$ are defined by the following system:

$$\begin{cases} a_{i,i+1} = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}, \\ b_{i,i+1} = -1, \\ c_{i,i+1} = y_i - \frac{y_{i+1} - y_i}{x_{i+1} - x_i} x_i. \end{cases} \quad (7)$$

According to these notations and Lemma 11, the segment bisector (B_i) related to the lines $(D_{i-1,i})$ and $(D_{i,i+1})$ is given by the following equation:

$$\frac{a_{i-1,i}x + b_{i-1,i}y + c_{i-1,i}}{\sqrt{a_{i-1,i}^2 + b_{i-1,i}^2}} = \pm \frac{a_{i,i+1}x + b_{i,i+1}y + c_{i,i+1}}{\sqrt{a_{i,i+1}^2 + b_{i,i+1}^2}}. \quad (8)$$

In Eq. (8), there is a choice of two possible cases according to the sign $-$ or $+$. For the case of skeletonization, the bisectors inside the angles are chosen as described in Fig. 13. To do this, the following condition is added where the middle of the segment $[M_{i-1}M_{i+1}]$ must belong to the mid-bisector B_i of the angle $\widehat{M_i} = \widehat{M_{i-1}M_{i+1}}$. Precisely, if $(\frac{y_{i-1}+y_{i+1}}{2}, \frac{x_{i-1}+x_{i+1}}{2})$ verifies Eq. (8) with $+$, then the case of equality-with $-$ is rejected, and vice versa.

It will be easy to have a suitable equation for B_i in the form $y = mx + n$; therefore, Eq. (8) becomes

$$y = A_{i-1,i,i+1}x + B_{i-1,i,i+1}, \quad (9)$$

where

$$\begin{cases} A_{i-1,i,i+1} = \frac{-a_{i-1,i}\sqrt{a_{i,i+1}^2 + b_{i,i+1}^2} \pm a_{i,i+1}\sqrt{a_{i-1,i}^2 + b_{i-1,i}^2}}{-b_{i-1,i}\sqrt{a_{i,i+1}^2 + b_{i,i+1}^2} \pm b_{i,i+1}\sqrt{a_{i-1,i}^2 + b_{i-1,i}^2}}, \\ B_{i-1,i,i+1} = \frac{-c_{i-1,i}\sqrt{a_{i,i+1}^2 + b_{i,i+1}^2} \pm c_{i,i+1}\sqrt{a_{i-1,i}^2 + b_{i-1,i}^2}}{-b_{i-1,i}\sqrt{a_{i,i+1}^2 + b_{i,i+1}^2} \pm b_{i,i+1}\sqrt{a_{i-1,i}^2 + b_{i-1,i}^2}}. \end{cases} \quad (10)$$

The indices $A_{i-1,i,i+1}$ and $B_{i-1,i,i+1}$ correspond to the points M_{i-1} , M_i , and M_{i+1} . Now, let the next incrementation follow the same procedure to determine the bisector segments of $(D_{i,i+1})$ and $(D_{i+1,i+2})$. It leads to the equation of the bisector $(B_{i+1}) : y = A_{i,i+1,i+2}x + B_{i,i+1,i+2}$, where $A_{i,i+1,i+2}$ and $B_{i,i+1,i+2}$ are computed using Eq. (8).

Once the bisectors (B_i) and (B_{i+1}) are computed, the internal intersections $N_{i,i+1}$ having the coordinates $(n_{i,i+1}, m_{i,i+1})$ are computed according to the following expression:

$$\begin{cases} n_{i,i+1} = B_{i,i+1,i+2} \frac{B_{i+1,i,i+2} - B_{i-1,i,i+1}}{A_{i,i+1,i+2} - A_{i-1,i,i+1}}, \\ m_{i,i+1} = A_{i-1,i,i+1}n_{i,i+1} + B_{i-1,i,i+1}. \end{cases} \quad (11)$$

The internal nodes $N_{i,i+1}$ are stored in a given set of internal nodes. The two bisector lines (B_i) and (B_{i+1}) are considered prioritized and deleted from the next process. They are regarded as beams of extinguished fire.

Test: If a segment bisector of an angle O_i , which does not meet any bisector, so it cannot be prioritized with anyone see Fig. 14. In this case, the node is calculated as follows:

The tangent T_i is computed at the angle O_i , then we consider the opposite segment C_i of the angle O_i in the border of X . Then, conduct the test as follows:

- If the two segments T_i and C_i are parallel, the node N_i will be the middle point between the point of the angle and C_i as described in Fig. 15.
- If the two segments T_i and C_i are not parallel, let's extrapolate the two segments and calculate their internal bisector R_i , then the node will be the intersection of R_i with B_i the bisector of the angle O_i (Fig. 16).

Then, the loop to the next angles is incremented and follows the same process.

The result leads to the identification of nodes N_i .

All the external nodes M_i and internal nodes N_i are stored in the set $PN = E \cup N$.

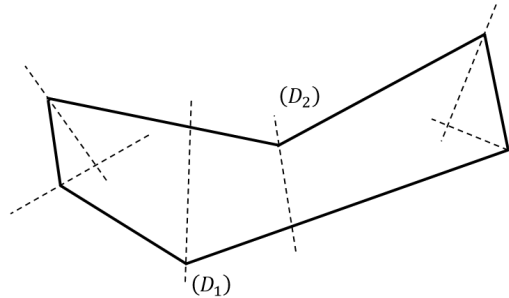


Fig. 14. The two bisectors D_1 and D_2 are not prioritized because they do not meet any bisector inside the polygon.

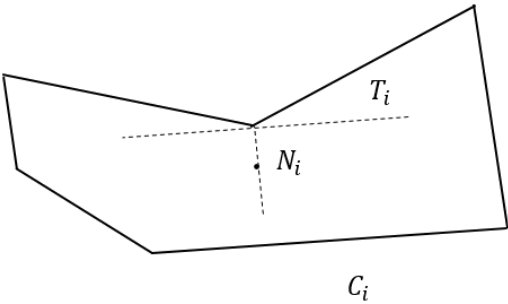


Fig. 15. The tangent at the angle T_i and the opposite segment of the angle O_i are parallel, we choose the middle inside P .

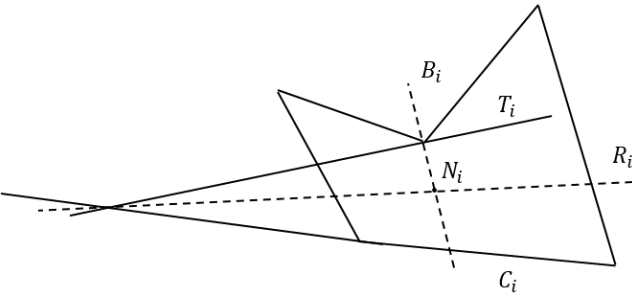


Fig. 16. The node is the intersection of B_i and R_i which is not necessary in the middle.

Stop Step II

Step III: Connecting the vertices of the skeleton

Once all the nodes are calculated and stored in PN, they are connected and to the opposite angles lower than 180° (Fig. 17). To conclude, the branches coming from angles greater than 180° are not considered (Fig. 17).

Stop Algorithm (Bisectors)

The network of segments linking all the nodes and their opposite angles lower than 180° represents the skeleton of the polygon P .

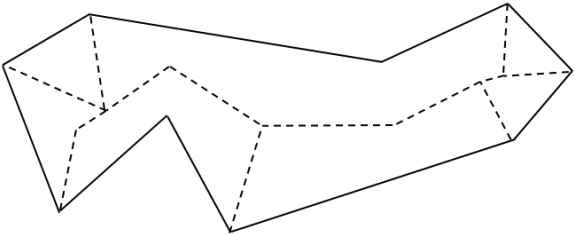


Fig. 17. The skeleton segments attach nodes to opposite angles lower than 180° .

5. Some Other Techniques

5.1. *Straight skeleton methods*²²

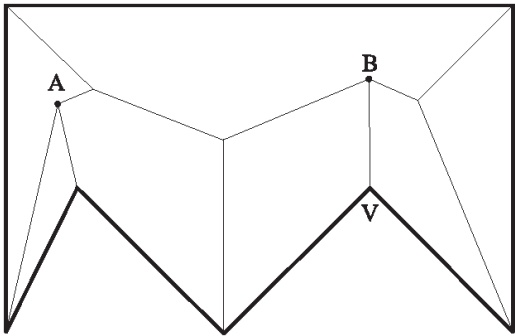


Fig. 18. The straight skeleton $\text{Sk}(P)$ of the input polygon P is defined by wavefronts emanating from P . The dashed arc is traced by a ghost vertex, see Ref. 22.

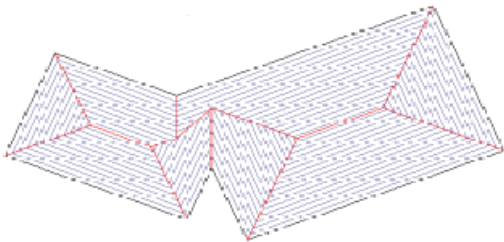


Fig. 19. A simple polygon where $\forall e f(e)$ (shaded) has a non-simple boundary. One wavefront is depicted by dotted lines. The bold edges have weight 3 and the two vertical edges that form the corridor have weight -1 . All other edges have weight 1. The arc between the two corridor edges geometrically coincides with other arcs. See Ref. 22.

5.2. *Bisectors technique: Implementation of the proposed algorithm*

The algorithm proposed in this paper is implemented on a Linux server running on a dell cluster (Dell VxRail S570", 10X2.5", Flash) using the Python language. The execution of the developed prototype program demonstrates that the algorithm proposed produced the results expected. However, to test the proposed algorithm, multiple basic and complex figures had been processed on the prototype system as a proof of concept. Furthermore, a comparison with the well-known and existing skeletonization function in "skimage.morphology" library has been discussed in the next section.

Certainly, the figures generated show the results of the skeletonization approach proposed in this paper. However, the quality of the skeleton is tightly dependent on

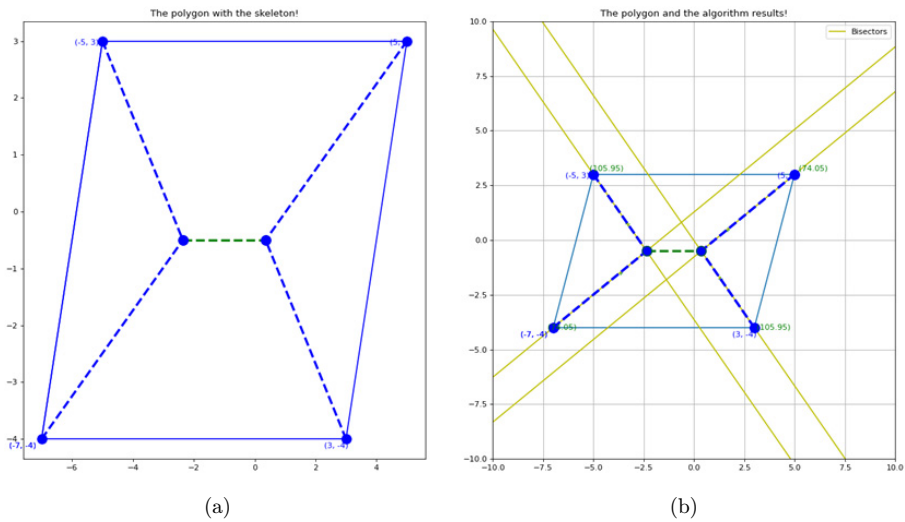


Fig. 20. The skeleton segments connecting nodes of a quadrilateral.

the Cartesianization rate, that is, the higher the border points selected, the more details are preserved and thus, the more representative the skeleton. However, a reasonable tradeoff has to be respected.

5.3. Method *skimage.morphology*

“*skimage.morphology*” is a Python library, that includes the *skeletonize* function, which, implements an approach that involves several passes of the image; in each

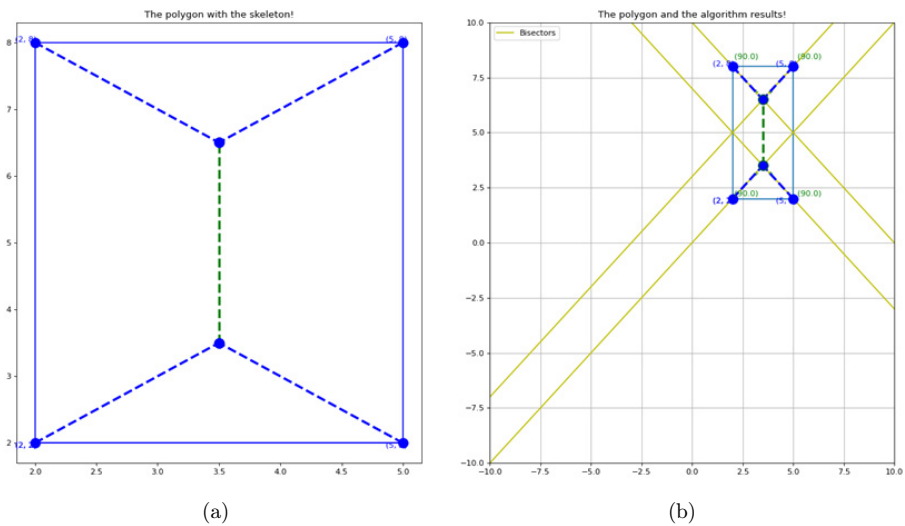


Fig. 21. The skeleton segments connecting nodes of a rectangle.

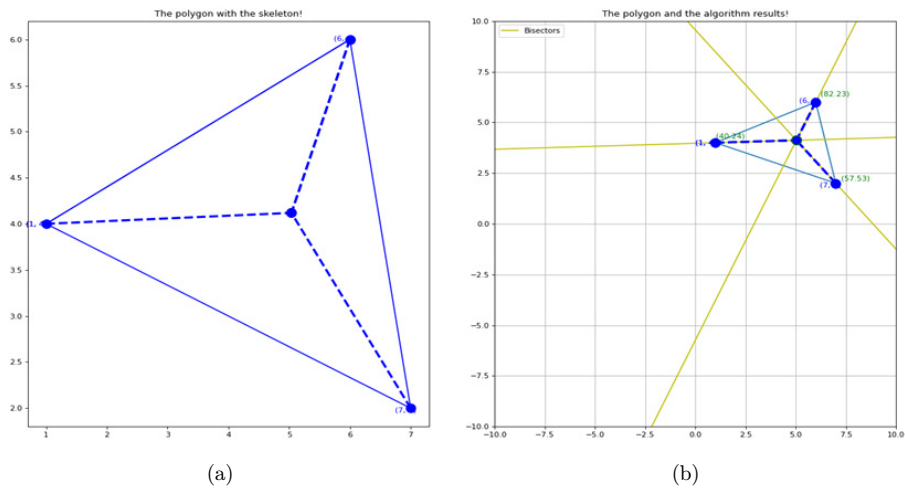


Fig. 22. The skeleton segments connecting nodes of a triangle.

pass, border pixels are discovered and eliminated, if they do not disrupt the connection of the relevant object. To compare, and shed light on the accuracy of both approaches the skeletonize function is applied to multiple shapes (inverted images) used in this research (Fig. 23). Furthermore, the resulting skeletons are superimposed, which reveals that both sets of skeletons are closely similar, however, some differences have been noticed as seen in Fig. 25.

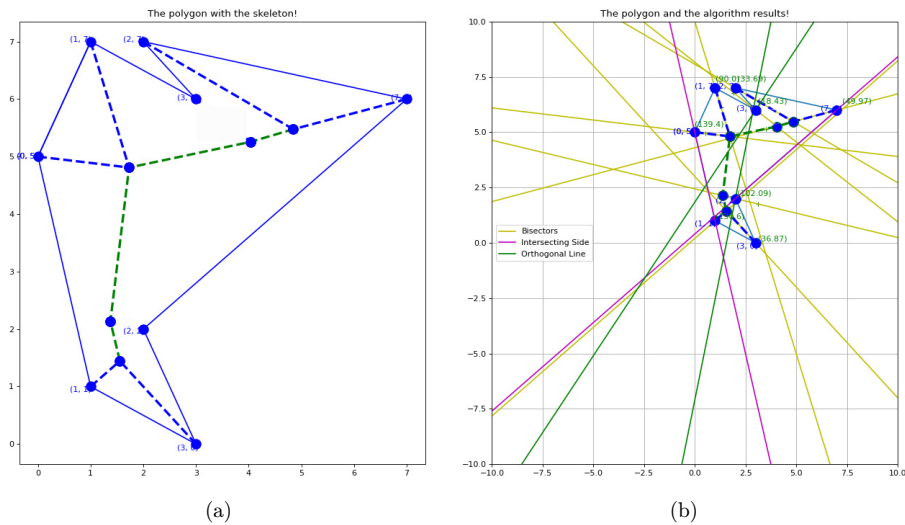


Fig. 23. The skeleton segments connecting nodes of polygon.

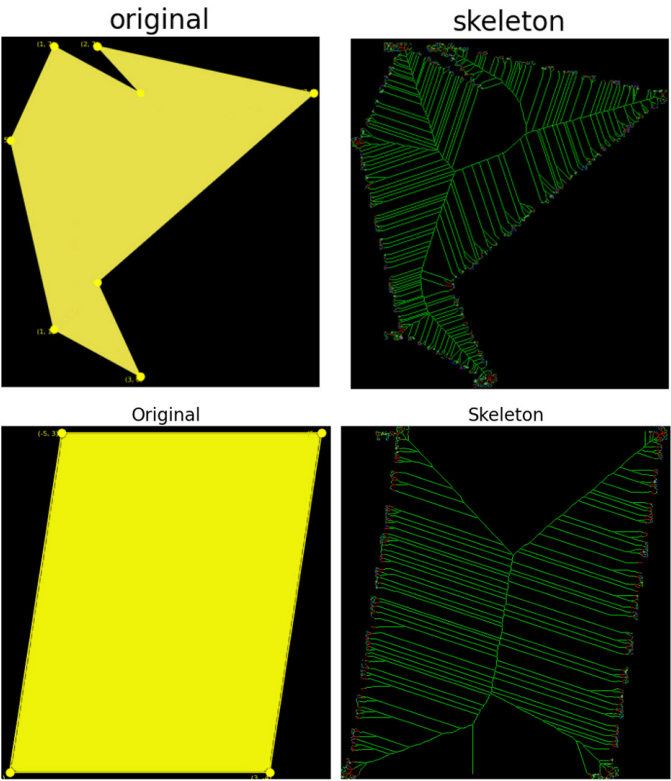


Fig. 24. The skeletons' segments connecting nodes of the polygon using the skeletonize() function.

Comparison and discussion of the selected methods

Certainly, the presented three techniques, straight skeleton,²³ bisectors technique proposed by this study, and Skimage morphology integrated into Python libraries have to be benchmarked concerning the mathematical reference of skeletonization, defined by the balls and maximum disk as stated in Definition 4.

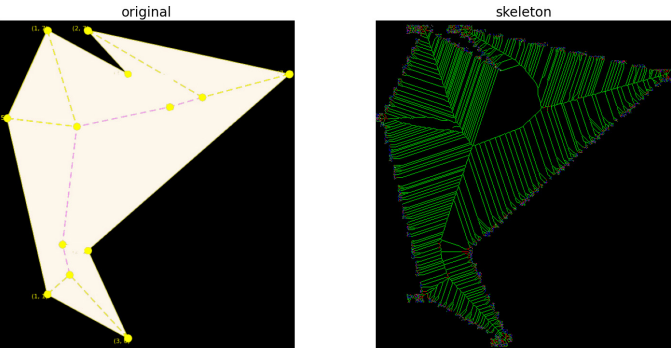


Fig. 25. The resulting skeletons of the same shape using both approaches.

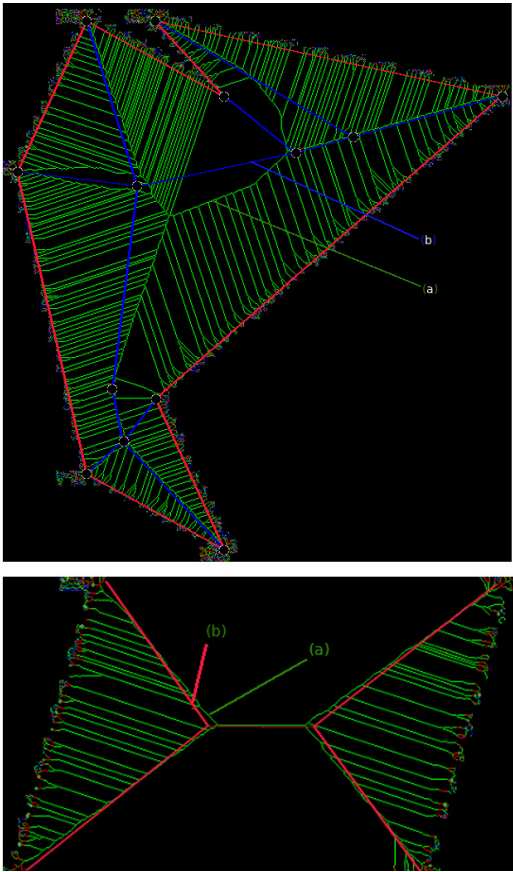


Fig. 26. Superimposition of skeletons generated by both methods (skeletonize() (a) and bisectors method (b)).

(1) For the straight skeleton method, it has a technical anomaly at the level of the branches of the skeleton attached to the acute angles greater than 180° , that is to say, the segment $[V, B]$ in Fig. 27 should not appear in the skeleton, because referring at the definition of maximum balls and disks, the only maximal disk tangent to V is

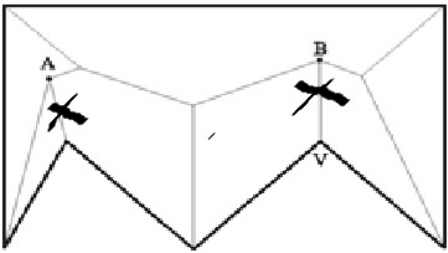


Fig. 27. Anomalies of straight skeleton method.

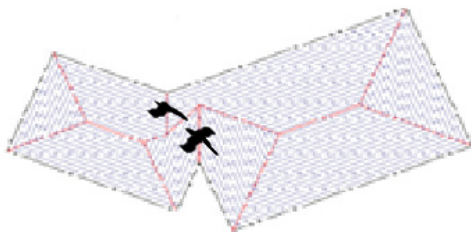


Fig. 28. Anomalies of straight skeleton method.

the disk with center B . Therefore, all the other points of the segment $[V, B]$ cannot represent centers of maximal disks, which disqualify the whole segment to be in the skeleton of Fig. 27.

Furthermore, Fig. 28 also exhibits the same anomaly, that is, the segments from angles greater than 180° should not appear in the skeleton.

(2) The technique (2) of the bisectors proposed in this paper respects the definition of the maximum disks. Figure 21 shows that the bisector method excludes segments from angles greater than 180° . The second advantage is that the proposed method uses the angle of the exposed points of the shape to extract the skeleton. For example, to calculate the skeleton of a rectangle, the four exposed points of the rectangle are used, to calculate the skeleton of a polygon with N sides, the N angles are used, that is to say, N bisectors. This method minimizes computation time and increases precision according to the most fundamental one.

(3) The method of Skimage morphology respects well the mathematical definition of maximal disks. The only drawback is that it relies on a large number of points on the boundary to extract the skeleton. As we observe in Fig. 26, a set of straight lines perpendicular to the boundary is used to retrieve the skeleton while our method uses four straight bisectors.

(4) Comparison of the Voronoi technique and the proposed technique.

First of all, we must emphasize that the Voronoi VSk skeletonization method is based on the Voronoi VD diagram: The VSk consists of first determining the list of boundary points of the object, this one which leads to an approximate polygon of the object. Then, Voronoi uses the VD diagram to determine a series of segments attached inside the object this segment connection then determines the medial axis called VMA. The question that arises for this technique is the following: At what level is this medial axis equivalent to a set of all centers of the maximal disks? It is clear that the Voronoi skeleton (medial axis) is very connected according to the Euclidean metric which allows a good image of the 2D object on a one-dimensional (1D) object. See Ref. 24. Our proposed method follows almost the same approach at the beginning of the VSk technique, this is what we call the Cartesianization of the shape to determine the exposed boundary points, this is for the polygonization of the shape. However, the second part (the proposed skeletonization) is based on the

angles formed by the border points. Once we have fixed the vertex angles of the object, we use the bisectors to determine the interior nodes. By tying these knots, we come across exactly the Voronoi segments that form the VSk.

Conclusion: Both the VSk and the proposed method determine the medial axis from the boundary points of the polygon, except the approach is different because Voronoi determines internal segments and the proposed method determines internal node points.

So, it is eventually clear to notice that the proposed method should be faster because it gives us output node points inside the object while VSk gives us output segments inside the object. It is known that the calculation time of a point is less than the calculation time of a segment (one segment = two points).

This brief comparison suggests that the proposed method in this paper surpassed the other methods, either in accuracy, performance, or both. The execution time for both methods is shown in Table 1.

Table 1.

Method	Sample exec time in seconds	Execution time s (avg: 100 runs)	Time diff.
Proposed method	0.289	0.293	
	0.296		
	⋮		
	0.295		
Skimage morphology	3.567	3.45	
	3.391		
	⋮		
	3.404		
System specifications			
Fabricant : Modèle : Évaluation : Processeur : Mémoire installée (RAM) : Type du système : Stylet et fonction tactile : Dell Precision 5510  Indice de performance Windows Intel(R) Core(TM) i7-6820HQ CPU @ 2.70GHz 2.70 GHz 16.0 Go (15.8 Go utilisable) Système d'exploitation 64 bits Saisie tactile et par stylet disponible avec 40 points de contact			

6. Conclusion

The outcomes of this study are based on several preliminary research studies on the Skeletonization of 2D and 3D geometric shapes, knowing that the general objective of this novel technique is to optimize the coverage and the control of a given region or shape. Mathematically, it is shown that the skeleton is the set of the closest points to the border of a corresponding region. Several methods concerning the extraction of the skeleton have already been published, as revealed by Felkel and Obdrzalek.²³ The added value of our bisector's method is to extract the skeleton, using the minimum number of exposed points, instead of the infinite points of the border and mitigate some issues of similar methods. This reduces the number of computer operations, and thus, the time it takes to execute the algorithm on a computing system. The details steps of this novel algorithm were presented and implemented using Python. In addition, a comparison between the proposed novel approach, the straight skeleton method, and the skeletonize function are included in the "skimage.morphology" Python library, which revealed that the proposed novel approach is more effective in generating skeletons and avoiding some limitations of similar methods such as adopting the bisector of external angles, as shown in Fig. 18.²² It should be noted that the publication following this paper will focus on the robustness, performance, and optimization of the proposed algorithm.

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